

Joontaek Oh

Postdoctoral Researcher, School of CS, UW-Madison, USA

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Google Scholar: scholar.google.com/citations?user=WKX0HNsAAAAJ

RESEARCH INTEREST

Operating System, Filesystem, Storage system, Flash Storage, Database systems, Manycore scalability

EDUCATION

Korea Advanced Institute of Science and Technologies (KAIST) <i>Ph.D in Electrical Engineering</i>	Daejeon, Korea <i>Mar. 2020 – Feb. 2024</i>
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Hanyang University <i>Ph.D student, Department of Computer Software</i>	Seoul, Korea <i>Sep. 2018 – Feb. 2020</i>
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Hanyang University <i>MS in Computer Science</i>	Seoul, Korea <i>Mar. 2016 – Aug. 2018</i>
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The Korean Academic Credit Bank System <i>BS in Information Security Engineering</i>	Seoul, Korea <i>Mar. 2013 – Aug. 2015</i>
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EXPERIENCE

Postdoctoral Researcher <i>University of Wisconsin-Madison</i>	Sep. 2024 – Present <i>WI, USA</i>
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Postdoctoral Researcher <i>Korea Advanced Institute of Science and Technologies (KAIST)</i>	Mar. 2024 – Aug. 2024 <i>Daejeon, Korea</i>
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PUBLICATIONS

Conferences

- [**ACM SOSP '26**] Seunghyun An, **Joontaek Oh**, Ming Liu, University of Wisconsin-Madison, “Scheduling Linux Threads under I/O Chiplet Wall Using cSwitch”, In Proc. of 23rd USENIX Symposium on Networked Systems Design and Implementation (SOSP) 2026, Sept. 29th - Oct. 2nd, 2026
- [**USENIX NSDI '26**] Chengdong Wang, **Joontaek Oh**, Ming Liu, “Switched or Switchless: An Empirical Study of SAN Architecture for Disaggregated Storage”, In Proc. of 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI) 2026, May. 4-6, 2026
- [**ACM SAC '25**] Juwon Kim, Dongeon Kim, Seung Won Yoo, Myoengin Cheon, **Joontaek Oh**, and Youjip Won, “lwFSCK: Light-weight Filesystem Check”, ACM Symposium on Applied Computing (SAC) 2025, March 31 - April 4, 2025

4. [USENIX FAST '25] Jieun Kim, **Joontaek Oh**, Juwon Kim, Seung Won Yoo, and Youjip Won, "OPIMQ: Order Preserving IO stack for Multi-Queue Block Device", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2025, Feb. 25-27, 2025
5. [USENIX FAST '25] Juwon Kim, Seungjae Lee, **Joontaek Oh**, Dongkun Shin, and Youjip Won, "D2FS: Device-Driven Filesystem Garbage Collection", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2025, Feb. 25-27, 2025
6. [USENIX FAST '25] Seung Won Yoo, **Joontaek Oh**, Myoengin Cheon, Bonmoo Koo, Wonseob Jeong, Hyunsub Song, Hyeonho Song, Donghun Lee, and Youjip Won, "DJFS : Directory-Granularity Filesystem Journaling for Memory-Semantic SSD", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2025, Feb. 25-27, 2025
7. [IEEE NVMSA '24] Seungho Lim, Seung Won Yoo, **Joontaek Oh**, Wonseob Jeong, Hyunsub Song, Hyeonho Song, Donghun Lee, and Youjip Won, "Key-Space Partitioned LSM Tree for CMM-H", In Proc. of IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA) 2024, Aug 21 – 23, 2024
8. [USENIX FAST '23] **Joontaek Oh**, Seung Won Yoo, Hojin Nam, Changwoo Min and Youjip Won, "CJFS : Concurrent Journaling for Better Scalability", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2023, Feb, 20-23, 2022
9. [USENIX ATC '22] Juwon Kim, Minsu Jang, Danish Muhammad Teeshen, **Joontaek Oh**, and Youjip Won "IPLFS: Log-Structured File System without Garbage Collection", In Proc. of USENIX Annual Technical Conference (ATC) 2022, July. 11-13, 2022
10. [ACM SYSTOR '22] Seung Won Yoo, **Joontaek Oh**, and Youjip Won "O-AFA: Order Preserving All Flash Array", in Proc. of The ACM International Systems and Storage Conference (SYSTOR), Haifa, Israel, June. 13-15, 2022
11. [USENIX FAST '22] Dohyun Kim, Kwangwon Min, **Joontaek Oh**, and Youjip Won "ScaleXFS: Getting scalability of XFS back on the ring", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2022, Feb, 22-24, 2022
12. [USENIX FAST '22] **Joontaek Oh**, Sion Ji, Yongjin Kim, and Youjip Won, "exF2FS: Transaction Support in Log-Structured Filesystem", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2022, Feb, 22-24, 2022
13. [ICISS '19] Myeongseon Kim, **Joontaek Oh**, Youjip won, "Barrier enabled QEMU", In Proc. of ICISS 2019, Tokyo, Japan, Mar. 2019
14. [USENIX FAST '18] Youjip Won, Jaemin Jung, Gyeongyeol Choi, **Joontaek Oh**, Seongbae Son, Jooyoung Hwang, Sangyeun Cho "Barrier Enabled IO Stack for Flash Storage", in proc. of USENIX Conference on File and Storage Technologies (FAST), Oakland, CA, USA, Feb. 12-15, 2018 (Awarded Best Paper)

Journals

1. [ACM TOS] Youjip Won, **Joontaek Oh**, Jaemin Jung, Gyeongyeol Choi, Seongbae Son, Jooyoung Hwang, Sangyeun Cho "Bringing Order to Chaos: Barrier-Enabled I/O Stack for Flash Storage", ACM Transactions on Storage (TOS)
2. [ACM TOS] Jinsoo Yoo, **Joontaek Oh**, Seongjin Lee, Youjip Won, Jin-Yong Ha, Jongsung Lee, Junseok Shim, "OrcFS: Orchestrated File System for Flash Storage", ACM Transactions on Storage (TOS), Vol. 14, Issue 2, Apr, 2018

Posters and Workshops

1. [ACM HotNet '25] Seunghyun An, **Joontaek Oh**, Ming Liu, "Server Chiplet Networking", In Proc. of 24th ACM Workshop on Hot Topics in Networks (HotNets), 2025
2. [ACM APSys '21] Kyoungcho Koo, **Joontaek Oh**, Kwangwon Min, Youngjin Kwon, Youjip Won, "C2J: Compulsory Compound Transaction for Journaling Filesystem", In Proc. of ACM SIGOPS Asia-Pacific Workshop on Systems (APSys), 2021

3. [USENIX FAST '19] **Joontaek Oh**, Wonjong Lee, Youjip Won, “xF2FS: Supporting Multi-File Transaction in Log-Structured Filesystem”, In Proc. of 17th USENIX Conference on File and Storage Technologies (FAST), 2019
4. [IEEE NVMSA '18] **Joontaek Oh**, and Youjip Won. “Embedded DBMS Design for In-Vehicle Information Management.” 2018 IEEE 7th Non-Volatile Memory Systems and Applications Symposium (NVMSA). IEEE, 2018.
5. [USENIX FAST '16] Jinsoo Yoo, **Joontaek Oh**, and Youjip Won. “Preserving Bi-Modal Utilization for Segment Cleaning in Modern Log-Structured Filesystem”, In Proc. of 14th USENIX Conference on File and Storage Technologies (FAST), 2016

Patents

1. Youjip Won and **Joontaek Oh**, ”Commit block structure and device, for multiple file transaction”, US12346223B2, July 2025
2. Youjip Won and **Joontaek Oh**, ”On-disk data structure for commit and method of commit with the data structure in log-structured filesystem”, KR20220074807A, June 2022
3. Youjip Won and **Joontaek Oh**, ”Method to solve the problem that file operation is blocked because of journal conflict”, KR20220074804A, June 2022
4. Youjip Won and **Joontaek Oh**, ”Method and in-memory structure for a file operation processing multiple files”, KR20220074806A, June 2022
5. Youjip Won and **Joontaek Oh**, ”Method and apparatus for sending barrier command using dummy IO request”, KR20190096838A, Aug. 2019
6. Youjip Won and **Joontaek Oh**, ”Method and apparatus for parallel journaling using conflict page list”, KR20190096837A, Aug. 2019

PROJECTS

High-Performance Exabyte Storage Systems Samsung Electronics	May 2022 – May 2024
SNU-SKH Solution Research Center SK Hynix	Sep. 2021 – Aug. 2026
Disaggregated Memory System for Data Center Samsung Electronics	July 2021 – June 2021
Virtualization on Embedded Architecture Samsung Electronics	Jan. 2021 – Dec. 2021
Lock-free Scalable IO Subsystem Design NRF	Jan. 2020 – Dec. 2025
Optimization of Filesystem and DBMS for Data Center Naver	June 2019 – May 2020
Barrier-Enabled IO Stack for NVMe SSD SK Hynix	Nov. 2018 – Oct. 2019
Future Scalable OS IITP	June 2018 – May 2023
Scalable IO Stack for future storage system NRF	June 2017 – Mar. 2020
System Software for Byte Addressable NVM KEIT/MOTIE	Mar. 2016 – May 2017

AWARDS AND HONORS

Best Ph.D. student Award EE Dept., KAIST	Apr. 2023
Best TA Award EE Dept., KAIST	Oct. 2020
Best Paper Award USENIX FAST 2018	Feb. 2018

TECHNICAL SKILLS

Languages: C/C++, Python, SQL (MySQL, SQLite), R

Developer Tools: gcc/g++, gdb, Git, QEMU, gnuplot, vim, Emacs

System knowledge: In-depth knowledge in Linux kernel and EXT4, F2FS, XFS, Ceph, Linux-MD, etc.

INVITED TALKS

File Systems and I/O Stacks for Emerging Storage Devices

Gyeongsang National University, Jun. 12. 2025

Separating the Ordering Guarantee and Durability Guarantee in Modern IO Stack

Georgia Institute of Technology, Apr. 24. 2024

IO Stack Design for Better Scalability

IBM Research, May. 23. 2023

SERVICE

Journal Reviewer, ACM Transactions on Storage (TOS).

Sep. 2025

TEACHING

TA, Programming Structure for EE (EE 209)

Korea Advanced Institute of Science and Technologies (KAIST)

Mar. 2021 – June 2021

Daejeon, Korea

TA, Introduction to Operating Systems (EE 415)

Korea Advanced Institute of Science and Technologies (KAIST)

Sep. 2020 – Dec. 2020

Daejeon, Korea

TA, Commissioned Education of IO Subsystem

Samsung Advanced Technology Training Institute

July 2020

Suwon-si, Korea

TA, Unix Kernel Design (EE 488)

Korea Advanced Institute of Science and Technologies (KAIST)

Mar. 2020 – June 2020

Daejeon, Korea

TA, SK Hynix - KAIST ASK Program

Korea Advanced Institute of Science and Technologies (KAIST)

Feb. 2020

Daejeon, Korea

TA, Introduction to Operating Systems (EE 415)

Korea Advanced Institute of Science and Technologies (KAIST)

Sep. 2019 – Dec. 2019

Daejeon, Korea

TA, Commissioned Education of IO Subsystem

Samsung Advanced Technology Training Institute

June 2019

Suwon-si, Korea

TA, Unix Kernel Design (EE 488)

Korea Advanced Institute of Science and Technologies (KAIST)

Mar. 2019 – June 2019

Daejeon, Korea

TA, System Programming (CSE 4009)

Hanyang University

Sep. 2018 – Dec. 2018

Seoul, Korea

TA, Commissioned Education of IO Subsystem <i>Samsung Advanced Technology Training Institute</i>	Aug. 2018 <i>Suwon-si, Korea</i>
TA, Operating System (ELE 3021) <i>Hanyang University</i>	Mar. 2018 – June 2018 <i>Seoul, Korea</i>
TA, System Programming (CSE 4009) <i>Hanyang University</i>	Sep. 2017 – Dec. 2017 <i>Seoul, Korea</i>
TA, Operating Systems & System Programming (ITE 2032) <i>Hanyang University</i>	Sep. 2016 – Dec. 2016 <i>Seoul, Korea</i>
TA, Commissioned Education of Operating System <i>Samsung Electronics</i>	Aug. 2016 <i>Suwon-si, Korea</i>
TA, Introduction to Operating System (ELE 3021) <i>Hanyang University</i>	Mar. 2016 – June 2016 <i>Seoul, Korea</i>

REFERENCES

Prof. Youjip Won ywon@kaist.ac.kr	Dept. of Electrical Engineering, KAIST
Prof. Youngjin Kwon yjkwon@kaist.ac.kr	School of Computing, KAIST
Dr. Changwoo Min changwoo@igalia.com	Igalia
Prof. Ming Liu mgliu@cs.wisc.edu	Dept. of CS, UW-Madison
Prof. Andrea Arpaci-Dusseau dusseau@cs.wisc.edu	Dept. of CS, UW-Madison
Prof. Remzi Arpaci-Dusseau remzi@cs.wisc.edu	Dept. of CS, UW-Madison